



Probability with Engineering Applications: Week11

Piao Chen, ZJUI, Fall 2025

Programme

- Law of Large Numbers
- Central Limit Theorem
- Random sample, statistical model, and estimators

Law of Large Numbers

Expected Value

Suppose $X_1, \dots, X_{50}$ are independent random variables and that each $X_i \sim N(100, 4)$. The expected value of the average $\bar{X}_{50}$ is equal to

A. 50×100

B. 400

C. 200

D. 100

Variance

Suppose $X_1, \dots, X_{100}$ are independent random variables and suppose that each $X_i \sim N(20, 4)$. The variance of the average $\bar{X}_{100}$ is equal to

A. 0.04

B. 0.2

C. 0.02

D. 4

Melting points

Suppose $Y_1, \dots, Y_{100}$ are independent random variables with $E[Y_i] = m$ and $\text{Var}(Y_i) = 4 = 2^2$.

Let $\bar{Y}_{100} = \frac{1}{100} \sum_{i=1}^{100} Y_i$, the average.

Using Chebyshev's inequality find p such that $P(|\bar{Y}_{100} - m| > 1) < p$.

A. 4

B. $\frac{2}{5}$

C. $\frac{1}{25}$

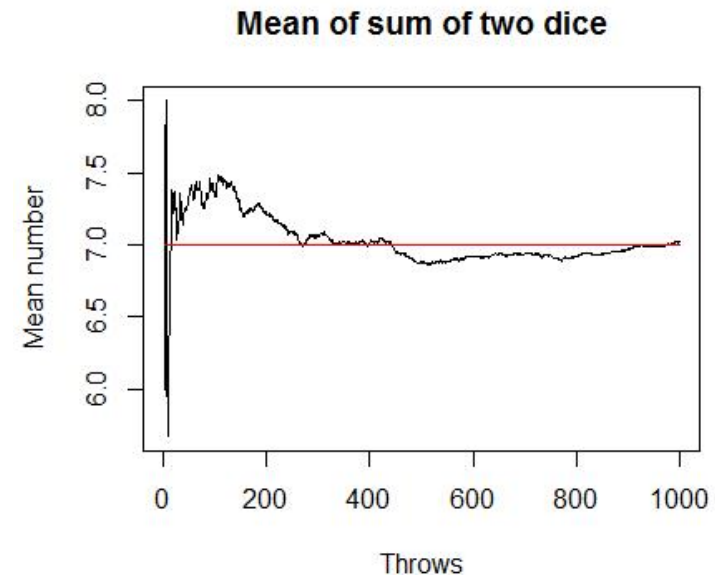
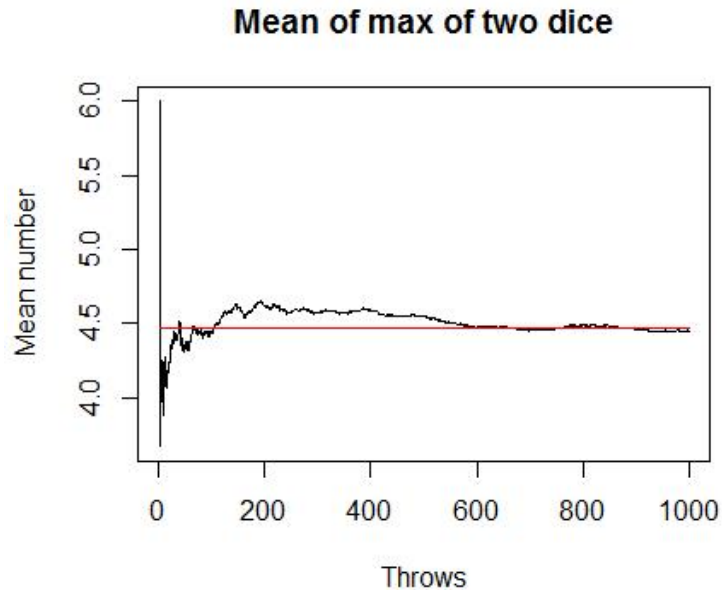
D. $\frac{1}{100}$

Law of Large Numbers

Theorem:

Let $X_1, \dots, X_n$ be an i.i.d. sequence with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$.
Then, for any $\epsilon > 0$:

$$P(|\bar{X}_n - \mu| > \epsilon) \rightarrow 0, \text{ as } n \rightarrow \infty.$$



Election polls

A fraction p of the population votes for party ABC. How to estimate p ?

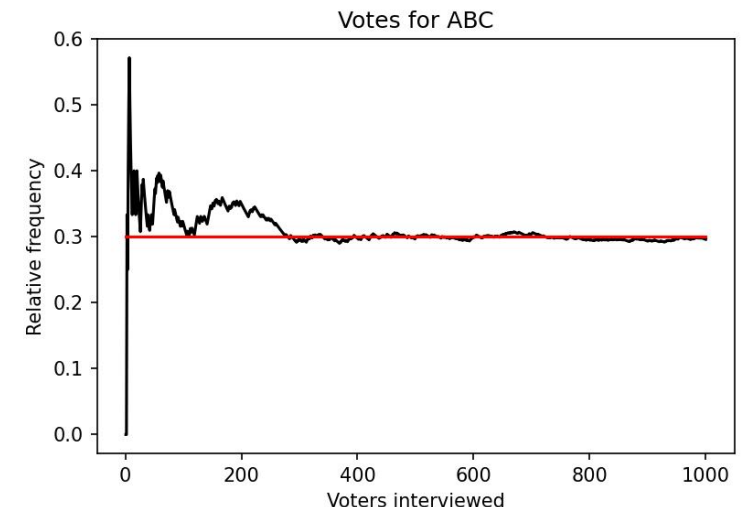
Pick n voters at random and let $X_i = \begin{cases} 1, & \text{if person } i \text{ votes for ABC,} \\ 0, & \text{otherwise.} \end{cases}$

Then $\bar{X}_n = \frac{\text{number of people voting for ABC}}{n} = \text{relative frequency}$

$E[X_i] = p$, so for any $\epsilon > 0$

$$P(|\bar{X}_n - p| > \epsilon) \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Interpretation: $\bar{X}_n \approx p$ for n large.



Practice now

Let $X_1, X_2, \dots$ be an independent sequence of $U(-1, 1)$ random variables and let $T_n = \frac{1}{n} \sum_{i=1}^n X_i^2$. Then there exists a real number a such that for all $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} P(|T_n - a| > \varepsilon) = 0.$$

Determine a and prove this statement.



Central Limit Theorem

A very fragile component



An industrial robot works non-stop and contains a very fragile component. Its active lifetime X , measured in days, is distributed as $\text{Exp}(1)$. Upon failure, the component is automatically replaced by a new one.

Suppose there are 100 components available, including the one that is currently used. What is the probability that the robot will run out of components in the next 90 days?

We need to know $P\left(\sum_{i=1}^{100} X_i < 90\right)$.

But we do not know the density function of $\sum_{i=1}^{100} X_i$.



Central Limit Theorem

Theorem:

Let $X_1, \dots, X_n$ be an i.i.d. sequence with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. For $n \geq 1$, let Z_n be defined by

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}.$$

Then for any number a it holds that

$$P(Z_n \leq a) \rightarrow P(Z \leq a), \text{ as } n \rightarrow \infty,$$

where Z has a $N(0, 1)$ distribution.



Central Limit Theorem

In other words: for large n , the variable Z_n approximately has a normal distribution:

$$Z_n \overset{d}{\approx} N(0, 1).$$

Since $\bar{X}_n = \frac{\sigma}{\sqrt{n}} Z_n + \mu$,

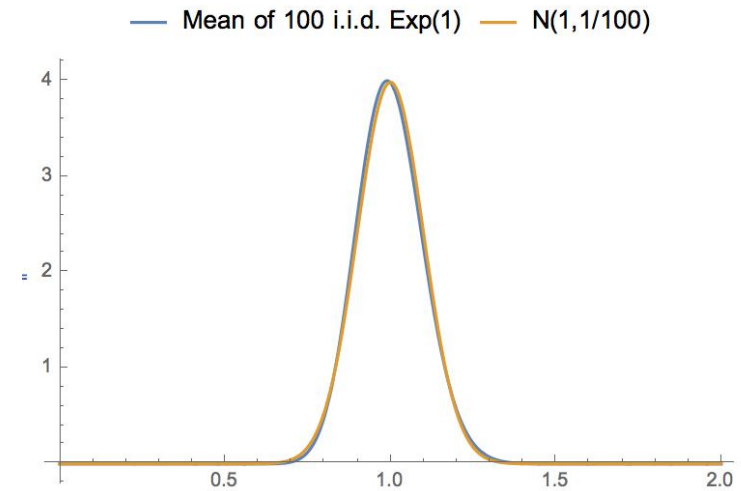
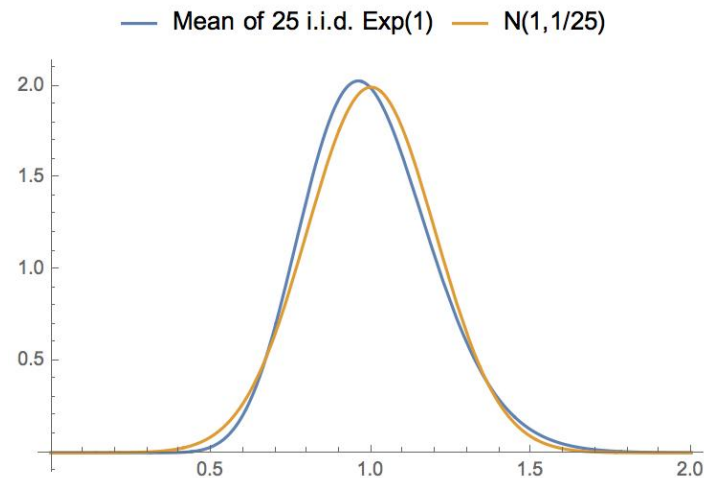
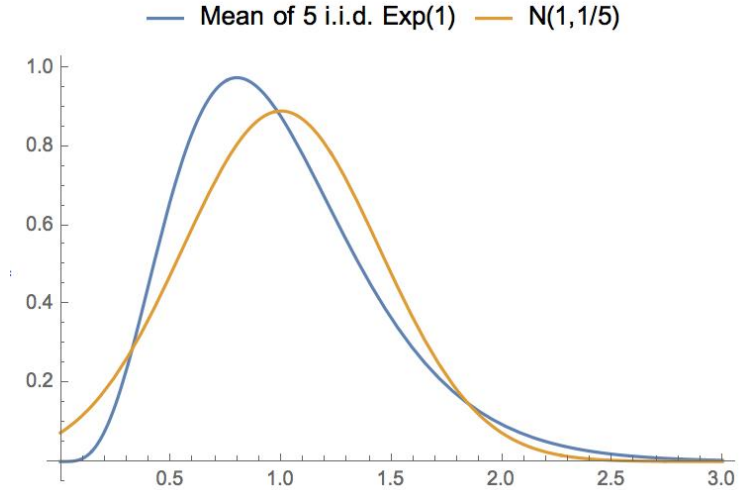
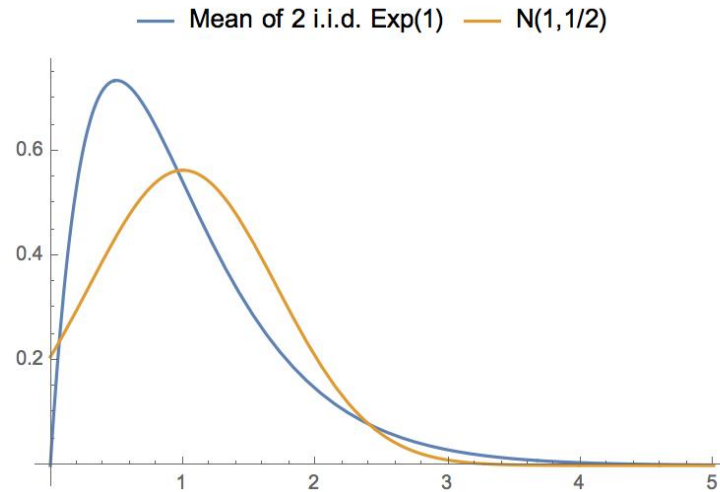
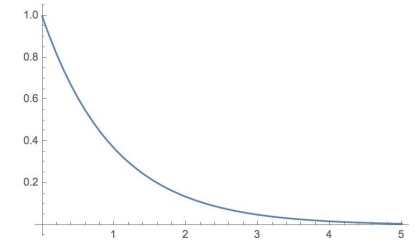
$$\bar{X}_n \overset{d}{\approx} N(\mu, \sigma^2/n).$$

And since $\sum_{i=1}^n X_i = n\bar{X}_n$,

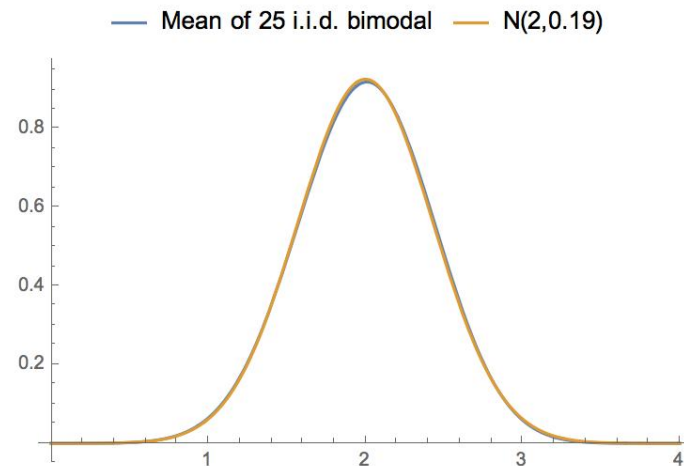
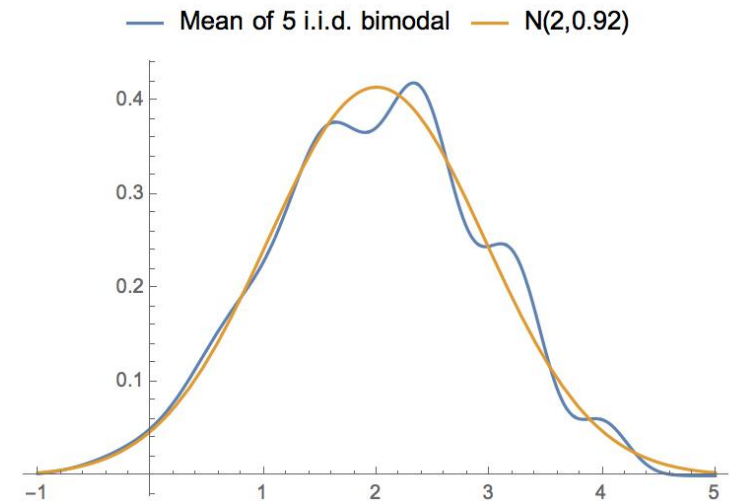
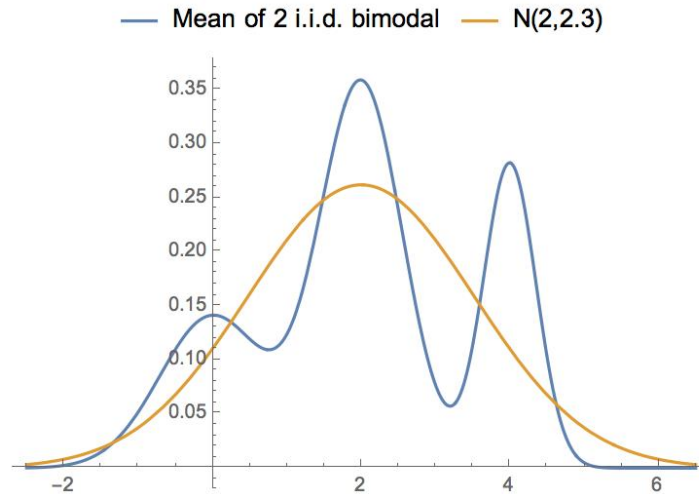
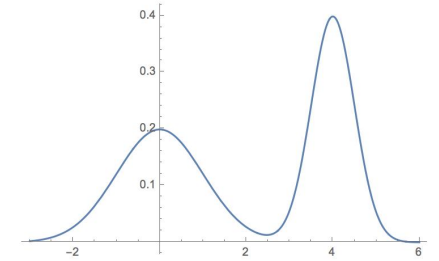
$$\sum_{i=1}^n X_i \overset{d}{\approx} N(n\mu, n\sigma^2).$$



Central Limit Theorem in pictures: skewed distribution



Central Limit Theorem in pictures: bimodal distribution



A very fragile component



Lifetime of a component: $X \sim \text{Exp}(1)$.

Suppose there are 100 components available, including the one that is currently used. What is the probability that the robot will run out of components in the next 90 days?

Since $E[X_i] = 1$ and $\text{Var}(X_i) = 1$, by the CLT we know that

$\sum_{i=1}^{100} X_i \stackrel{d}{\approx} N(100 \cdot 1, 100 \cdot 1)$. Hence

$$\begin{aligned} P\left(\sum_{i=1}^{100} X_i < 90\right) &\approx P(S < 90) = P\left(\frac{S - 100}{10} < \frac{90 - 100}{10}\right) = P(Z < -1) \\ &\approx 0.1587 \end{aligned}$$

where $S \sim N(100, 100)$ and $Z \sim N(0, 1)$.



Example

An astronomer wants to determine the distance to the moon, but changing atmospheric conditions result in mean 0, iid noise to be added to each distance measurement

The signal model is: $X_i = d + n_i$

- Measurement = (distance to the moon) + (iid noise)
- $E[n_i] = 0$
- $\text{Var}(n_i) = 4$

$$E[X_i] = E[d + n_i] = d \text{ and } \text{Var}(X_i) = \text{Var}(d + n_i) = \text{Var}(n_i) = 4$$

How many measurements are required to be 95% confident the distance estimate $\hat{d}$ is within ± 0.5 m?

Example

[illegible]

Sum of normal random variables

Sum of normal random variables

Theorem:

Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ be independent random variables. Then $X + Y$ is also a normal random variable:

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

Corollary:

Let $X_1, \dots, X_n$ be an i.i.d. sequence with $X_i \sim N(\mu, \sigma^2)$. Then

$$\sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2) \quad \text{and} \quad \bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$



Central limit theorem

Let $X_1, \dots, X_n$ be i.i.d. random variables with distribution function F_X
Consider the following statements:

1. If n is large, then the distribution of X will be approximately normal.
2. If n is large, then $\max\{X_1, \dots, X_n\}$ will be approximately normal.

Which of these statements is/are true?

- A. None
- B. Only 1
- C. Only 2
- D. Both

Exercise

Suppose $X_1, \dots, X_{64}$ are independent random variables and suppose that each $X_i \sim \text{Par}(3)$.

Compute/approximate $P(\sum_{i=1}^{64} X_i \leq 96)$.



Random sample and statistical model

Random sample and statistical model

Definition:

A random sample is a collection of RVs

$$X_1, X_2, \dots, X_n$$

that have the same probability distribution and that are independent.

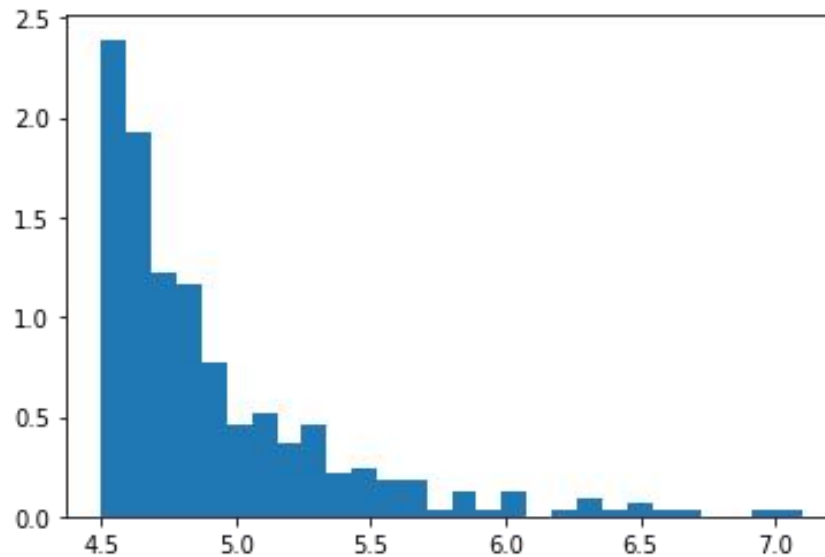
A dataset consisting of repeated measurements $x_1, x_2, \dots, x_n$ is modelled as the realization of a random sample $X_1, X_2, \dots, X_n$.

n is called the sample size.

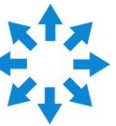


Earthquakes

351 earthquakes of magnitude at least 4.5 in North America, from 2011 to January 2022.



Source: United States Geological Survey (usgs.gov)

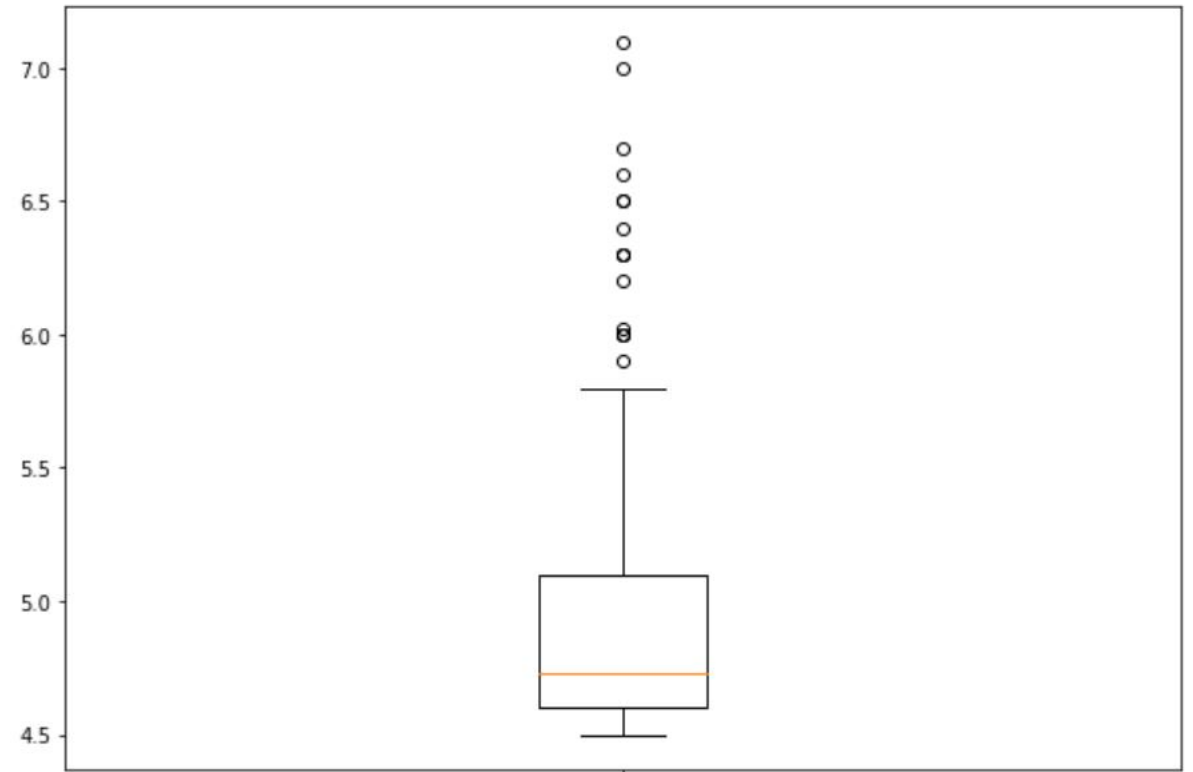


Earthquakes

4.5
5.4
4.5
4.7
4.58
4.75
4.6
4.7
4.9
4.5
4.84
6.2
5.65
4.6
4.7
4.5
4.8
5.1
...

The five-number summary:

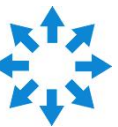
- Minimum: 4.5
- Lower quartile: 4.6
- Median: 4.73
- Upper quartile: 5.1
- Maximum: 7.1



Can we find a mechanism that mimics the data generating process?



Build a stochastic model for the data generating process.



Model distribution and sample statistics

Model distribution

Definition:

The probability distribution of each random variable from a random sample is called the model distribution.

Definition:

A random variable $h(X_1, \dots, X_n)$ depending only on the random sample $X_1, \dots, X_n$ is called a sample statistic.

Statistics:

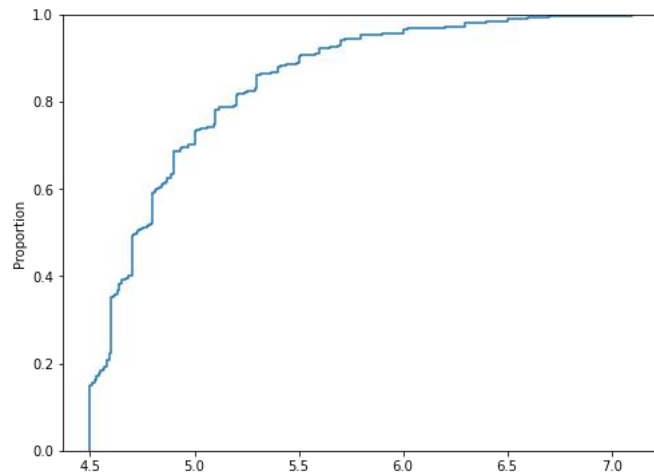
- Identify features (e.g. density, expectation) of the model distribution using sample statistics from the data set
- Which model distribution fits a particular dataset best?



Sample statistics vs. distribution features

1. Sample mean vs expectation
2. Empirical distribution function vs model distribution function
3. Histogram vs density of the model distribution
4. Sample median vs $F^{\text{inv}}(0.5)$

The expectation of the random variable is *approximately* 4.902.



$F^{\text{inv}}(0.5)$ is *approximately* equal to 4.73.



Assuming a distribution

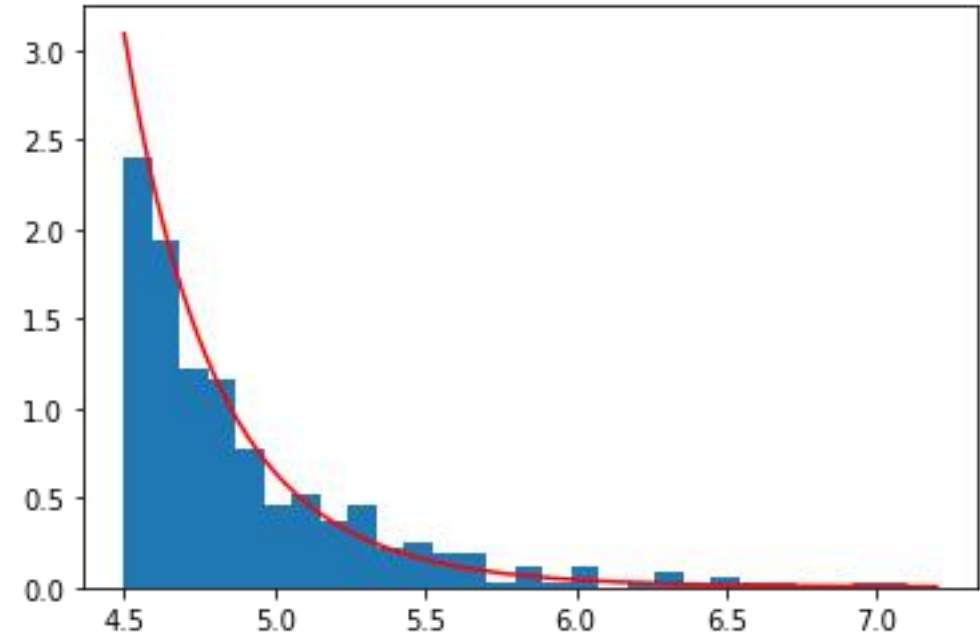
Earthquake magnitudes are usually Pareto distributed.
Since only magnitudes of at least 4.5 are given, we propose a *scaled* Pareto distribution:

$$f(x) = \frac{\alpha (4.5)^\alpha}{x^{\alpha+1}}.$$

The median of this distribution is $4.5 \cdot 2^{1/\alpha}$.

The sample median is 4.73.

Based on this, we estimate $\hat{\alpha} = 13.9$ (red).



Assuming a distribution

Earthquake magnitudes are usually Pareto distributed.
Since only magnitudes of at least 4.5 are given, we propose a *scaled* Pareto distribution:

$$f(x) = \frac{\alpha (4.5)^\alpha}{x^{\alpha+1}}.$$

The median of this distribution is $4.5 \cdot 2^{1/\alpha}$.

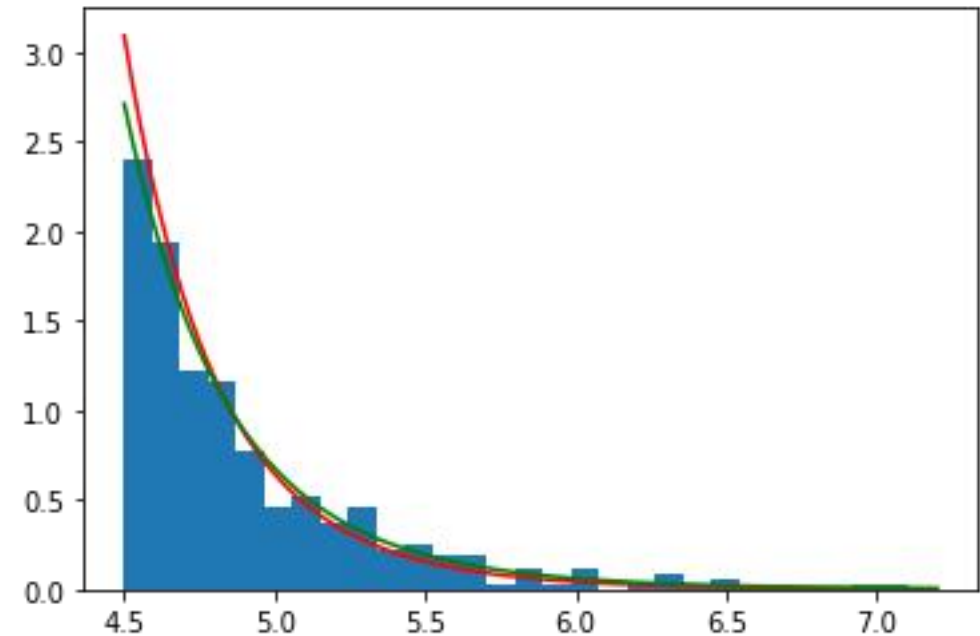
The sample median is 4.73.

Based on this, we estimate $\hat{\alpha} = 13.9$ (red).

The mean of this distribution is $4.5 \frac{\alpha}{\alpha-1}$.

The sample mean is 4.902.

Based on this, we estimate $\hat{\alpha} = 12.2$ (green).



Assuming a distribution

Earthquake magnitudes are usually Pareto distributed.
Since only magnitudes of at least 4.5 are given, we propose a *scaled* Pareto distribution:

$$F(x) = 1 - \left(\frac{4.5}{x} \right)^\alpha.$$

The median of this distribution is $4.5 \cdot 2^{1/\alpha}$.

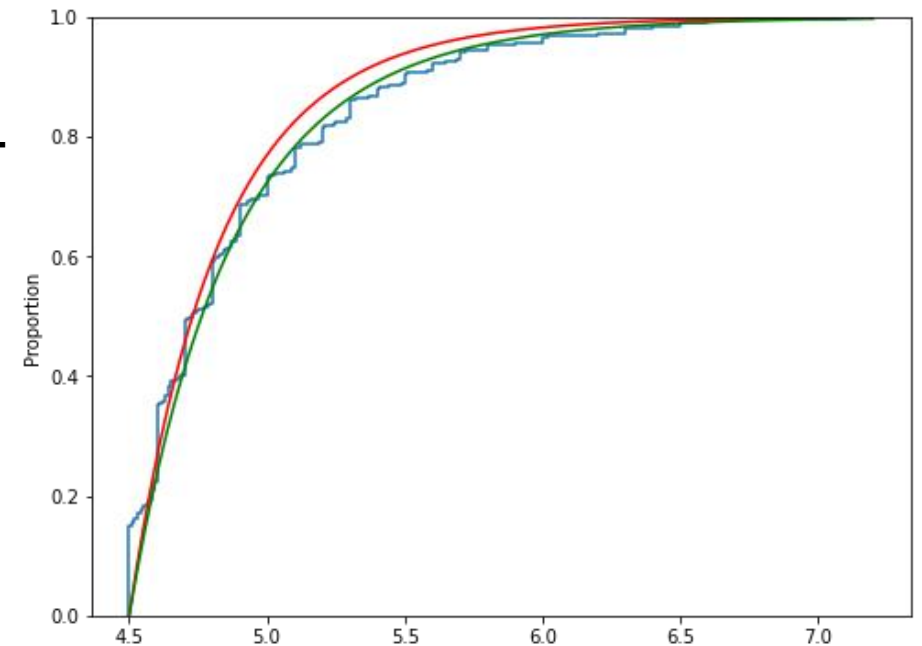
The sample median is 4.73.

Based on this, we estimate $\hat{\alpha} = 13.9$ (red).

The mean of this distribution is $4.5 \frac{\alpha}{\alpha-1}$.

The sample mean is 4.902.

Based on this, we estimate $\hat{\alpha} = 12.2$ (green).

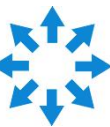


Estimators and bias

Drawing balls from an urn

An urn contains N balls, labeled $\{1, 2, \dots, N\}$. We draw three balls, with replacement.

How can we estimate N , based on this sample x_1, x_2, x_3 ?



Estimate and estimator

Definition:

A real value t that only depends on the dataset $x_1, x_2, \dots, x_n$, i.e.

$$t = h(x_1, x_2, \dots, x_n)$$

and that is used to estimate a fixed, unknown parameter θ , is called an estimate of θ .

Then t is a realization of a random variable

$$T = h(X_1, X_2, \dots, X_n)$$

which we call an estimator of θ .



Sampling Distribution

Definition:

Let $T = h(X_1, X_2, \dots, X_n)$ be an estimator based on a random sample $X_1, X_2, \dots, X_n$. The probability distribution of T is called the sampling distribution of T .



Bias

Definition:

Let T be an estimator for the parameter θ . The quantity $E[T] - \theta$ is called the **bias** of the estimator T . We say T is

- negatively biased if $E[T] - \theta < 0$;
- unbiased if $E[T] - \theta = 0$;
- positively biased if $E[T] - \theta > 0$.

Note: If T is unbiased then $E[T] = \theta$



Unbiased estimator

Suppose T is an estimator for the parameter θ with expectation $E[T] = 2\theta + 1$. Which of the following statements is true?

- A.** $S = 2T + 1$ is an unbiased estimator for θ
- B.** $S = 2T - 1$ is an unbiased estimator for θ
- C.** $S = \frac{1}{2}T - \frac{1}{2}$ is an unbiased estimator for θ
- D.** $S = \frac{1}{2}T - 1$ is an unbiased estimator for θ

Drawing balls from an urn

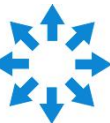
An urn contains N balls, labeled $\{1, 2, \dots, N\}$. We draw three balls, with replacement.

How can we estimate N , based on this sample x_1, x_2, x_3 ?

$$T_1 = X_1$$

$$T_2 = \bar{X}_3$$

$$T_3 = \max(X_1, X_2, X_3)$$





See you next week